Impact of Inventory Management Practices on MSMEs: A Regional Study in India

S Ramesh Murthy* and Padmalatha N A**

This study examines the inventory management (IM) practices, reasons, influencing factors and impact of IM on the performance of micro, small and medium enterprises (MSMEs). Primary data was collected using a survey questionnaire among 40+ MSMEs in Baddi Industrial Area, Himachal Pradesh, India. The study reveals that MSMEs generally follow poor IM practices and non-scientific forecasting methods, leading to capital being tied up due to excess inventory or stockouts, delayed delivery, customer dissatisfaction, reduced profits, operational inefficiencies, and diminished competitiveness. Factors such as limited awareness, lack of access to technology and finances, and a shortage of skilled manpower contribute to inadequate implementation of scientific IM practices. The study establishes that poor IM practices and unscientific forecasting practices have an impact on the inventory cost, which in tun impacts the firm's performance. To address these challenges, the study makes many recommendations.

Introduction

Inventories constitute a significant portion of the current assets in any business enterprise (Kruger, 2005). Lack of appropriate inventory management (IM) practices can give rise to a range of problems, including reduced productivity (Palange and Dhatrak, 2021), production of unwanted items, diminished customer commitment (Du *et al.*, 2015) accumulation of costly physical inventories, and frustration (Meyer, 1991). Hence, the cost savings resulting from improved IM practices are substantial (Meyer, 1991). Effective IM and control are crucial for the viability of a firm, as mismanagement of inventory can jeopardize its operations (Sprague and Wacker, 1996; and Holweg *et al.*, 2005). The management of inventories directly impacts a firm's financial strength and competitive position, as the approach taken towards IM affects working capital,

* Research Scholar, Dayananda Sagar University, Bengaluru, Karnataka, India; and is the corresponding author. E-mail: rameshshellm_72@yahoo.com

** Associate Professor, Dayananda Sagar University, Bengaluru, Karnataka, India. E-mail: drpadmalatha-socm@dsu.edu.in

© 2024 IUP. All Rights Reserved.

production, and customer service levels (Ng *et al.*, 1993; Vergin, 1998; and Sinha *et al.*, 2023). Inventory is also part of the routines and repeat routines (Saratchandra *et al.*, 2022) while acting as a driver of competitive advantage (Jones and Riley, 1985; and Bhatnagar and Teo, 2009). Therefore, it is imperative that organizations utilize scientific methods in the management of inventory whilst striving to do better than their competitors (Li *et al.*, 2006) in the challenging marketplace.

Micro small and medium enterprises (MSMEs) are defined in various ways across the globe. In India, a micro enterprise in the manufacturing sector is characterized by an investment in plant and machinery not exceeding ₹10 mn. A small enterprise has an investment in plant and machinery greater than ₹10 mn but less than ₹100 mn. A medium enterprise is defined as having an investment in plant and machinery greater than ₹100 mn but not exceeding ₹500 mn. MSMEs are the backbone of the Indian economy, contributing nearly 30% of the country's gross domestic product (GDP), generating employment opportunities for over 110 million people, particularly in rural and semi-urban areas where other job opportunities are limited. They are the largest employer after the agriculture sector and account for almost 50% of India's total exports. The employment opportunity provided by such firms is equally significant, for they lead to better living standards for young people who move from villages to towns and cities.

MSMEs are especially critical for emerging markets on account of their contribution to new employment generation, serving larger original equipment manufacturers (OEMs) and leading to the emergence of economic clusters that support larger industrial clusters. While clusters and the derived economics have been articulated in detail by Porter (1998), on-the ground evidence is manifest in the form of industrial clusters such as the automobile clusters around Chennai, India and Pune, India. Textile clusters have emerged in towns such as Tirupur and Ludhiana. The presence of larger OEMs along with the ancillary industries leads to a healthy self-sustaining industrial ecosystem. The Western world has several such ones—for instance, the auto design firms clustered around Turin, Italy or the automotive cluster in Detroit, US and the machine tools and manufacturing industrial clusters around Cincinnati, Ohio, US. All clusters have developed with a small number of large firms that are supported by several hundreds of smaller firms which are essentially MSMEs.

Despite the acknowledged significance of inventories for a firm's survival, IM practices in several MSMEs are very rudimentary and often use heuristic approaches (Widyastuti, 2018). In the formulation of business strategies, IM is often not considered a critical or strategic activity (Sprague and Wacker, 1996). There are fewer in-depth studies on the IM practices adopted by MSMEs in India. Therefore, this paper aims to fill this gap by examining the state of IM practices amongst MSMEs in the Baddi Industrial Area, Himachal Pradesh, India.

Study Setting

The state of Himachal Pradesh in India is known for tourism and industrial zones. From an industrial zone context, the Indian government set up specific departments that would facilitate and expedite industrial activities. One such body is the Development Commissioner Ministry of Micro Small & Medium Enterprises (DCMSME) (https://dcmsme.gov.in). The Baddi region in Solan District of Himachal Pradesh had drawn up an industrial profile of the district during 2014-15 (https://dcmsme.gov.in/old/dips/District%20Industrial%20 Potential%20Solan.pdf). The document outlines the topography as well as the industrial scenario prevailing in the district. The total geographical area of the district is 1,936 sq. km. Forest area accounts for 11% of the total area.

Baddi has about 3,000 companies that provide raw materials, intermediaries and finished goods to industries such as pharmaceuticals, automobiles, consumer goods, industrial machinery and food processing. Recent work in the district shows that specific zones have been demarcated for expanding the existing manufacturing cluster at Baddi (Sharma, 2023), indicating the resurgence of industrial activity. Baddi-Barotiwala-Nalagarh (BBN) is an industrial hub identified for development as part of infrastructural advancement of the Amritsar-Kolkata Industrial Corridor project. Within the ambit of this larger initiative, the BBN corridor has been developed by National Industrial Corridor Development Corporation (NICDC), a special purpose vehicle for continuous industrial development.

Theoretical Foundations and Literature Review

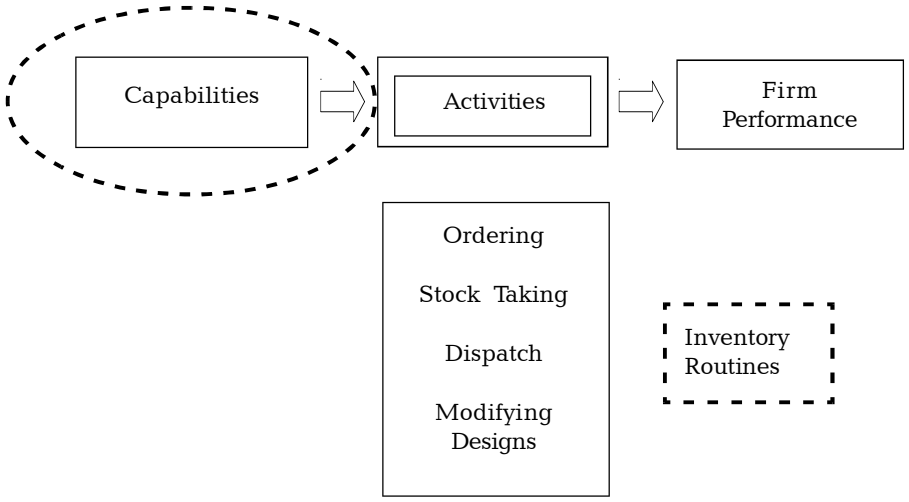
Certain practices adopted by MSMEs can be beneficial in establishing foundational aspects derived from specific theories. A starting point would be to draw inferences from the resource-based view (Wernerfelt, 1984; and Barney, 1991), wherein the scholars classify all firm resources into tangible and intangible ones. Firms are able to articulate their tangible resources, while struggling to take stock of their intangible ones. In the context of MSMEs, inventory can be said to be a tangible resource that acts as a bargaining point, helping organizations to leverage them for routine business transactions. Hence, inventory may act as an operational lever, enabling organizations to use it as a potent tool, despite its being a part of the repeat routines. The resource-based view laid more emphasis on the firms' being unable to identify the intangibles, whereas in most cases, they can easily list down the tangible resources.

Extending the applications from Porter's (1980) clear classification of a) Cost leadership, b) differentiation and c) focus as the three generic strategies which could lead firms to competitive advantage, Hill and Jones (2011) went on to propose and substantiate their theories on the building blocks of competitive advantage. Their exposition envisioned a) Cost efficiency, b) quality, c) innovation and d) customer responsiveness as the building blocks of competitive advantage. With regard to MSMEs, we can clearly see the application of inventory as a lever that could help firms achieve

cost efficiency, provided effective and scientific IM practices were used. Moreover, for scenarios which called for quick response during uncertain times, inventory buffers could also lead to superior customer responsiveness.

Since most of the MSMEs studied are suppliers to larger OEMs, their focus is directed towards capabilities and activities which would consequently drive towards firm performance. Their repeat routines also align them well with the structural framework proposed by Hill and Jones (2011); the construct can be portrayed in the form of a capabilities framework (Figure 1). Examples of activities could be any business process such as placing a purchase order or despatch against an order received.

Figure 1: The Capabilities Framework in Inventory Perspective



The dynamic capabilities perspective (Pisano and Teece, 1984) applied to MSMEs may be construed as a quest for constant improvement in the capabilities of the smaller firms since their demands are volatile and ever-changing. Moreover, since the bargaining power rests with the buyer ecosystem, the smaller firms which tend to be part of the MSME universe need to constantly upgrade their capabilities. On the other hand, a lot of their activities (Figure 1) remain routine, including the activities undertaken as part of IM practices. The dotted line depicting the capabilities ellipse indicates that capabilities have to be constantly refined and honed; firms cannot choose to be complacent.

At the turn of the century, Lawson and Samson (2001) strongly observed that within the umbrella of capabilities, firms need to provide for innovation; they opined that the innovation capabilities need to be enhanced with a view to serving their customer

interests. More recent studies emphasized that MSMEs need to innovate using technology (Gupta and Barua, 2016), while constantly striving to be ahead of competing firms in capabilities (Khurana *et al.*, 2019).

IM has significance for any enterprise because effective practice of IM allows an enterprise to minimize inventory costs and avoid shortage or excess of material resources, leading to cost efficiencies within the firm. Eloranta and Raisanen (1988); and Demberere and Kasongo (2021) argued that poor quality of forecasts is one of the factors contributing to shortage or excess inventory in MSMEs. Chikan (1990) observed that a sound IM system is a decisive factor in a firm's success that may over time act as a key building block of competitive advantage. Natarajan (1991) discussed the relationship between IM and competitive advantage, bringing into focus factors such as cost, delivery and quality; he argued that inventory costs are determined not only by their level of inventory but also by the time the materials spend in the system. Chikan and Whybark (1990) observed that MSMEs are slow to adopt and implement contemporary IM practices. MSMEs are often satisfied with archaic, non-scientific practices such as end-of-year stocktaking, trading and maintaining profit and loss accounts, even though many benefits can be derived from implementing a formal inventory management system. Given that many SMEs lack professional expertise (Kumar *et al.*, 2022) and generally take decisions based on intuition and elementary IM practices (Mohanty, 1985), investments in inventory are not always costed accurately or appropriately. Peel *et al.* (2000) revealed that small firms tend to have a relatively high proportion of current assets, less liquidity, exhibit volatile cash flows, and a high reliance on short-term debt. Similarly, Howorth and Westhead (2003) suggested that small companies tend to focus on some areas of IM where they can expect to improve marginal returns. For small businesses, an efficient IM is a vital component of success and survival (Muchaendepi *et al.*, 2019). Grablowsky (1976) showed a significant relationship between various success measures and the adoption of IM policies. Given the peculiarities of small businesses, Peel and Wilson (1996) stressed the need for efficient inventory management.

There are many factors that cause IM problems in small scale industries. Chan *et al.*, 2017 argued that underproduction, overproduction, stockout situation, delays in the delivery of raw materials, planning, documentation/store records, knowledge of employee/staff skill and funding are some of the reasons for IM-related problems. Rajeev (2008) found that use of heuristics in IM, low emphasis placed on forecasting techniques, random ordering of materials, low levels of training and development, and lesser support from computerized methods as well as low importance given to purchasing and variable lead-time lead to IM problems in the context of machine tool enterprises in Bengaluru, India. Ahmad and Zabri (2016) demonstrated that systematic approach of IM was only utilized by 33% of MSMEs surveyed. The rule of thumb was the most popular IM techniques among respondents; this indicates that convenience and judgement played a role in critical inventory decisions. Owner-managers' attitude

and knowledge in IM have significant influence on IM practice. IM and managerial competence are significantly associated with financial performance of small businesses (Orobia *et al.*, 2020). Lack of knowledge of benefits and competitive advantage results in one fourth of the MSME not implementing any kind of IM practice (Mangla *et al.*, 2018). Apart from technology, factors like human resource, economic condition, organizational culture of SMEs are vital in improving the performance of inventory management. The study indicated that despite being aware of IM practice, most of them base their decision on heuristics (Narayanapillai, 2013).

Several studies have established the positive impact of IM on the performance of the organization (Ahmad and Zabri, 2018). Superior IM practices can lead to enhanced competitive advantage and improved organizational performance (Atnafu and Balda, 2018). IM leads to economy and efficiency, nurtures competitive ability and paves the way for cost reduction (Chalotra, 2013). MSMEs that implemented IM practices, which included reflected frequent stock verification and EOQ-based raw material ordering, achieved lower inventory cost and higher inventory turnover rate (Mangla *et al.*, 2018). Kaewchura (2021) established that inventory control systems had the greatest influence on competitive advantage, followed by inventory control practices and information technology. Narayanapillai (2013) found that variables such as safety stock planning, employee training and better IM practices differentiate the performance of firms from the rest; Buteau (2021) noted that digital technologies could significantly accelerate the growth of MSMEs in India. Therefore, we find that lack of formalized and scientific methods in inventory management leads to increased inventory costs for the firm.

Forecasting and Inventory Cost/Performance

While some studies have looked at digital interventions in the performance of MSMEs (Bhatti *et al.*, 2022), the authors agree that absence of rigorous IM practices has diluted the potential performance; Kumar and Chandra (2023) proposed a hybrid framework to mitigate some prevailing IM problems in the context of pharmaceutical MSMEs, while noting that arcane forecasting methods and frequent intervention of 'experts' led to unstructured decision making. Syntetos *et al.* (2010) discussed the impact of subjectivity observed in forecasting methods as they find that such approaches may end up harming the accuracy of scientific forecasting. The findings suggest that managerial intervention on account of process-specific knowledge may hamper the implementation of rigorous, software-based methods. Upadhye *et al.* (2010) discussed that lean management principles could be used to improve IM practices; they argued that poor quality forecasts contribute to shortage or excess of inventory in MSMEs.

Hypotheses

H_1: *Poor IM practices increase the inventory costs (decrease the performance)*

H_2: *Poor forecasting methods increase the inventory cost (decrease the performance)*

Data and Methodology

A structured questionnaire (see Appendix) containing qualitative questions and questions based on a 5-point Likert scale was used to collect data from of 40+ MSME firms in Baddi industrial area.

The questionnaire focussed on collecting data on following points – inventory counting frequency, reordering frequency, inventory carrying costs, safety stock, impact of IM systems, production planning, lead time, physical inventory counting, inventory tracking tools, stockout prevention strategies, raw material sourcing, and substitute material usage during stockouts.

Descriptive analysis was done to establish if the IM practices and forecasting methods are good or poor. The study uses factor analysis to group the observed variables into 3 latent factors, namely, inventory practices, forecasting challenges and inventory cost. It then uses structural equation model to establish the causal relationship among the latent factors and test the hypothesis that poor IM practices and archaic forecasting method negatively impact the inventory costs, which is used as an indicator to measure the performance and competitiveness of the firm.

Reliability of the Instrument

The reliability of the questionnaire was measured using Cronbach's alpha. A coefficient of 0.9850 suggests that the items in the entire set measure the same underlying construct with very high consistency and reliability.

Findings

Descriptive Analysis

The findings highlight the key aspects of IM practices in small-scale industries in the Baddi industrial area of Himachal Pradesh, India. Most surveyed firms (58%) conduct weekly inventory counts, while 28% perform daily counts, and 15% carry out monthly counts. On average, 10 stock-keeping units (SKUs) are counted during inventory. Firms conducting daily inventory counts reorder stock after 6 days, those conducting weekly counts reorder after 8 days, and firms performing monthly counts reorder after 30 days. Firms conducting daily counts maintain safety inventory, while those conducting monthly counts do not, resulting in higher inventory levels and carrying costs for the latter group. Production planning strategies vary among the small-scale industries surveyed. Around 36% plan production for fast-moving SKUs on a make-to-stock basis, while 28% plan production based on customer orders. Excel is the preferred inventory management tool for 60% of the firms, while 33% rely on accounting systems like Tally.

To prevent stockouts, firms employ different strategies. Approximately 26% focus on regular inventory counts and monitoring, while 45% prioritize employee training

and empowerment. Some firms believe monitoring the entire supply chain is crucial for avoiding stockouts. Regarding sourcing practices, 75% of the firms prefer direct sourcing from manufacturers but face challenges such as high minimum order quantities and advance payment requirements. Intermediaries are often used, causing supply chain delays. Sales forecast based on past customer orders are used by 33% of firms, while another 33% collaborate with suppliers for sourcing raw materials. In the event of stockouts, firms borrow materials from friendly manufacturers (50%), arrange rush delivery (5%), wait for stocks to become available (15%), or use substitute materials (30%). The surveyed small-scale industries do not adopt theoretical concepts like economic order quantity (EOQ) and reorder point (RoP), inventory classification methods like ABC Analysis, or inventory optimization techniques or software. Limited evidence of supply chain collaboration with suppliers and customers is also observed.

Based on the descriptive analysis, it is concluded that the MSMEs in Baddi industrial area follow poor inventory management practices and non-scientific/rudimentary/ rule of thumb forecasting methods.

Factor Analysis

Based on the Kaiser-Meyer-Olkin test results, the overall measure of sample adequacy (MSA) value is 0.628. It can be concluded that the variables in the dataset are suitable for factor analysis.

Principal component analysis (PCA) with varimax rotation was first carried out using JMP-SAS software. The FPCA analysis is shown in Table 1.

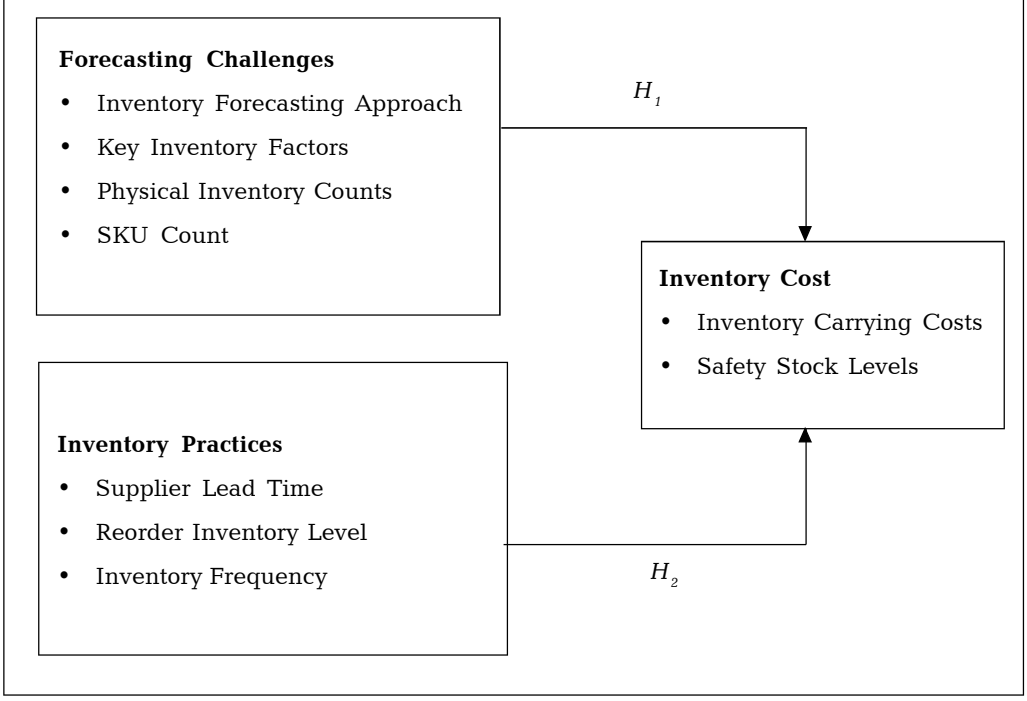
Table 1: FPCA Analysis Results

Variable	Loading		
	Factor 1	Factor 2	Factor 3
Supplier Lead Time	0.99875		
Reorder Inventory Level	0.97984		
Inventory Frequency	0.52068		
Inventory Forecasting Approach		0.63641	
Key Inventory Factors		0.59277	
Physical Inventory Counts		0.49217	
Baddi Area Inventory Challenges		–0.4748	
SKU Count		0.38184	
High Inventory Levels		–0.0613	
Reasonable Carrying Costs			0.37647
Safety Stock Levels			0.3511

Table 1 shows the presence of 3 factors. The first category (Variables 1-3) was categorized as inventory management practices. The second category (Variables 4-8) was classified as forecasting challenges, and the third group (Variables 9-11) was categorized as inventory costs.

Figure 2 represents the conceptual framework of the study.

Figure 2: Conceptual Framework



Structural Equation Model and Hypothesis Testing

From Figure 3 and Table 2, it is established that the *p*-value of 0.0066 indicates statistical significance, suggesting a significant relationship between poor inventory practices and increase in inventory costs. The RMSEA value of 0.1282 indicates a good fit for the structural equation model. The Comparative Fit Index (CFI) value of 0.7777 is reasonable for a good fit. Hence, H_1 is accepted.

Figure 3: Output of Structural Equation Model

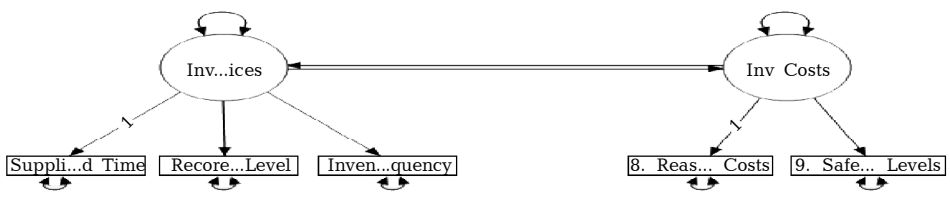


Table 2: Hypothesis Testing (H_1)					
Hypothesis	**Relationship**	***p*-Value**	**CFI**	**RMSEA**	**Decision**
H_1	Poor Inv Practices→ High Inv Costs	0.0066	0.7777	0.1282	Accepted

Regarding H_2, from the output of Figure 4 and Table 3, it is established that the *p*-value of 0.0001 indicates statistical significance, suggesting a strong relationship between poor forecasting practices and increase in inventory costs. But the RMSEA value of 0.2587 suggests a poor fit for the structural equation model and the CFI value of 0.1389 is below the threshold for a good fit. Therefore, the overall fit of the structural equation model is not strong, and the hypothesis is rejected.

Figure 4: Output of Structural Equation Model

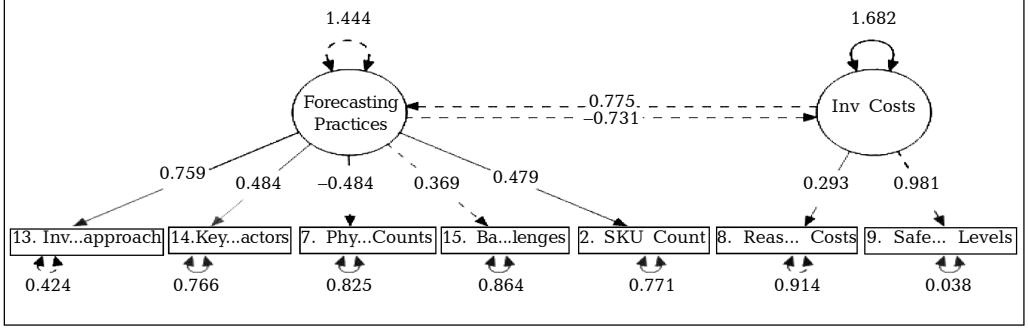


Table 3: Hypothesis Testing (H_2)					
Hypothesis	**Relationship**	***p*-Value**	**CFI**	**RMSEA**	**Decision**
H_2	Poor Forecasting Practices →High Inv Costs	0.0001	0.1389	0.2587	Rejected

Discussion

The descriptive analysis revealed that the forecasting methods employed by the MSMEs were rudimentary in nature. There were instances of excess inventory or stockouts. However, the results obtained from the structural equation model did not support the notion that poor forecasting methods directly translate into increased inventory costs. To further explore this discrepancy, additional data analysis was performed, and follow-up telephonic discussions were conducted with the respondents. It was discovered that the MSMEs deal with a relatively low number of SKUs. Their end customers are known, they receive annual orders with scheduled shipments, and they rarely encounter sudden orders from unknown customers. As a result, they are able to forecast their inventory needs with a reasonable level of accuracy. However, due to credit terms and

small minimum order quantities (MoQs), these MSMEs are compelled to rely on intermediaries for their supplies. The inconsistency in supply from intermediaries, coupled with sudden changes in prices, forces the MSMEs to avoid holding excessive inventory. Consequently, the observed instances of excess inventory or stockouts are primarily attributed to the behavior of suppliers rather than the forecasting practices of the MSMEs. Therefore, the rejection of H_2, which suggests a direct link between poor forecasting and inventory issues, is justified.

Conclusion

A comprehensive study conducted on 40 MSMEs in the Baddi industrial area revealed a significant deficiency in IM practices. These enterprises exhibited poor utilization of formal IM practices, characterized by a lack of integration between physical stock and accounting systems. The primary contributors to subpar IM practices were identified as the absence of a cohesive approach, a lack of awareness about IM among entrepreneurs, and a shortage of qualified personnel. Furthermore, various other factors compounded the situation, including working capital constraints, inadequate training and development, stagnation in human resource development, absence of an ERP system and software utilization, and organizational characteristics specific to MSMEs.

The study discovered the absence of formal inventory ordering policies, such as fixed quantity or fixed period ordering policies, in the surveyed SMEs. Instead, a haphazard procurement policy prevailed, indicating a random approach to material procurement. Moreover, reliance on imprecise rule-of-thumb methods for IM, limited emphasis on accurate forecasting, ad hoc material ordering practices, low utilization of computer systems, and a diminished focus on purchasing and variable lead-time were identified.

It was observed that MSMEs did not engage in collaborative purchasing from manufacturers, which could provide them with bargaining power and other benefits such as reduced lead time, lower minimum order quantity, credit facilities, and quantity discounts. The utilization of computers was limited to word processing and Microsoft Excel, with minimal focus on improving employee skills through training and development programs. Human resource development was found to be lacking.

The absence of effective IM practices renders organizations uncompetitive and inefficient, and negatively impacts profitability. Firms often face issues of excess inventory or stockouts, resulting in inventory discrepancies or incorrect products being available at the wrong time and place. This leads to high capital costs or an inability to meet demand, resulting in customer dissatisfaction and long-term consequences for the firm's survival.

Implications: MSMEs are the backbone of the Indian economy, and their competitiveness and profitability are crucial for national growth. It is imperative that MSMEs adopt effective IM strategies to minimize costs, enhance profitability, and

gain a competitive advantage. The government should play a vital role in supporting MSMEs by providing training, mentorship, and exposure to the benefits of modern IM practices. This includes implementing theoretical inventory control concepts like EOQ, ROP, and just-in-time (JIT) systems and utilizing inventory forecasting techniques like trend analysis, moving average, and exponential smoothing. Additionally, MSMEs can benefit from commercial-grade IM software such as Zoho inventory, which enables streamlined inventory tracking, stock monitoring, and insightful reporting. Measuring inventory performance through metrics like inventory turnover ratio and days inventory cover, and employing inventory classification methods such as ABC analysis, FSN analysis, and VED analysis, can help MSMEs prioritize their inventory effectively. Adopting conceptual concepts such as inventory optimization techniques, implementing inventory management software, and fostering supply chain collaboration with suppliers and customers can improve inventory management practices in the studied MSMEs. In addition to the theoretical concepts and conceptual concepts mentioned, there are a few other factors that impact the IM practices of MSMEs like the type of product or service being sold, the size of the business, the industry in which the business operates, the level of competition and the economic climate.

It is important for MSMEs to consider all these factors when developing and implementing inventory management practices. By embracing these solutions, small-scale industries can optimize their inventory, reduce costs, and improve operational efficiency overall.

Limitations and Future Scope: India is a vast country, and the government is setting up industrial clusters in various parts of the country. MSMEs in various clusters have different cultural and social factors. They face challenges which are unique to them. Hence, further studies focusing on different MSME clusters are necessary to understand their IM practices and reasons for not implementing modern IM practices. There is a need for research to understand the impact of lack of resources, inadequate infrastructure, and limited access to finance faced by MSMEs and their impact on IM practices. Future studies should also be done to explore the impact of cultural and social factors, the role of government policies, and the impact of technological advancements on IM practices in Indian MSMEs.

This study has several limitations. Firstly, the questionnaire items used to measure constructs were not standardized, highlighting the need for future research to revalidate measurement scales with more representative observations. Additionally, relying on a single respondent from a firm to assess the overall aspect of inventory management practice of that firm may not accurately reflect the reality as no single person is solely responsible for inventory within a firm. Therefore, future studies should aim to incorporate multiple respondents from each participating enterprise to enhance the research findings. Furthermore, the data collection from a single MSME cluster restricts the ability to generalize the results to the entire sector at a national level, and the findings may not be applicable to medium- and large-scale manufacturing firms.❖

References

1. Ahmad K and Zabri S M (2016), "Inventory Management Practices Among Malaysian Micro Retailing Enterprises", *Journal of Business and Retail Management Research*, Vol. 11, No. 1, pp. 103-115.

2. Ahmad K and Zabri S M (2018), "The Mediating Effect of Knowledge of Inventory Management in the Relationship Between Inventory Management Practices and Performance: The Case of Micro Retailing Enterprises", *Journal of Business and Retail Management Research*, Vol. 12, No. 2, pp. 83-93.

3. Atnafu D and Balda A (2018), "The Impact of Inventory Management Practice on Firms' Competitiveness and Organizational Performance: Empirical Evidence from Micro and Small Enterprises in Ethiopia", *Cogent Business & Management*, Vol. 5, No. 1, p. 1503219.

4. Barney J (1991), "Special Theory Forum the Resource-Based Model of the Firm: Origins, Implications, and Prospects", *Journal of Management*, Vol. 17, No. 1, pp. 97-98.

5. Bhatnagar R and Teo C C (2009), "Role of Logistics in Enhancing Competitive Advantage: A Value Chain Framework for Global Supply Chains", *International Journal of Physical Distribution & Logistics Management*, Vol. 39, No. 3, pp. 202-226.

6. Bhatti S H, Ahmed A, Ferraris A *et al.* (2022), "Big Data Analytics Capabilities and MSME Innovation and Performance: A Double Mediation Model of Digital Platform and Network Capabilities", *Annals of Operations Research*. https://doi.org/10.1007/s10479-022-05002-w, pp. 1-24.

7. Buteau S (2021), "Roadmap for Digital Technology to Foster India's MSME Ecosystem—Opportunities and Challenges", *CSI Transactions on ICT*, Vol. 9, No. 4, pp. 233-244.

8. Chalotra V (2013), "Inventory Management and Small Firms Growth: An Analytical Study in Supply Chain", *Vision*, Vol. 17, No. 3, pp. 213-222.

9. Chan S W, Tasmin R, Aziati A N *et al.* (2017), "Factors Influencing the Effectiveness of Inventory Management in Manufacturing SMEs", *IOP Conference Series: Materials Science and Engineering,* Vol. 226, No. 1, p. 012024. IOP Publishing.

10. Chikan A (1990), "Characterization of Production-Inventory Systems in the Hungarian Industry", *Engineering Costs and Production Economics*, Vol. 18, No. 3, pp. 285-292.

11. Chikán A and Whybark D C (1990), "Cross-National Comparison of Production-Inventory Management Practices", *Engineering Costs and Production Economics*, Vol. 19, Nos. 1-3, pp. 149-156.

12. Demberere J and Kasongo R M (2021), "The Effectiveness of Supply Chain Management Practices on Manufacturing Micro, Small and Medium Enterprises

(MSME) in Lusaka: A Case of Mandevu and Kalingalinga Markets", *Global Journal of Purchasing and Procurement Management*, Vol. 1, No. 1, pp. 1-21. https://doi.org/10.1016/j.ijpe.2015.02.013.

13. Du J, Zhang J and Hua G (2015), "Pricing and Inventory Management in the Presence of Strategic Customers with Risk Preference and Decreasing Value", *International Journal of Production Economics*, Vol. 164, pp. 160-166.

14. Eloranta E and Raisanen J (1988), "Delivery Performance In Small-To-Medium-Scale Industry", *S. Production and Inventory Management Journal*, Vol. 29, No. 1, p. 50.

15. Gupta H and Barua M K (2016), "Identifying Enablers of Technological Innovation for Indian MSMEs Using Best-Worst Multi Criteria Decision Making Method", *Technological Forecasting and Social Change*, Vol. 107, pp. 69-79.

16. Hill C W and Jones G R (2011), *Essentials of Strategic Management*, Cengage Learning.

17. Holweg M, Disney S, Holmström J and Småros J (2005), "Supply Chain Collaboration: Making Sense of the Strategy Continuum", *European Management Journal*, Vol. 23, No. 2, pp. 170-181.

18. Howorth C and Westhead P (2003), "The Focus of Working Capital Management in UK Small Firms", *Management Accounting Research*, Vol. 14, No. 2, pp. 94-111.

19. Kaewchura P (2021), "Role of Inventory Management on Competitive Advantage of Small and Medium Companies in Thailand", *Turkish Journal of Computer and Mathematics Education (TURCOMAT)*, Vol. 12, No. 8, pp. 2753-2759.

20. Khurana S, Haleem A and Mannan B (2019), "Determinants for Integration of Sustainability with Innovation for Indian Manufacturing Enterprises: Empirical Evidence in MSMEs", *Journal of Cleaner Production*, Vol. 229, pp. 374-386.

21. Kruger G A (2005), "A Statistician Looks at Inventory Management", *Quality Progress*, Vol. 38, No. 2, p. 36.

22. Kumar D and Chandra D (2023), "A Hybrid Framework to Model Resilience in the Generic Medicine Supply Chain of MSMEs", *Benchmarking: An International Journal*. Vol. 30, No. 6, pp. 2189-2224. https://doi.org/10.1080/09537287.2022.2131620

23. Kumar M, Raut R D, Mangla S K *et al.* (2022), "The Adoption of Artificial Intelligence Powered Workforce Management for Effective Revenue Growth of Micro, Small, and Medium Scale Enterprises (MSMEs)", *Production Planning & Control*, pp. 1-17.

24. Lawson B and Samson D (2001), "Developing Innovation Capability in Organisations: A Dynamic Capabilities Approach", *International Journal of Innovation Management*, Vol. 5, No. 03, pp. 377-400.

25. Li S, Ragu-Nathan B, Ragu-Nathan T S and Rao S S (2006), "The Impact of Supply Chain Management Practices on Competitive Advantage and Organizational Performance", *Omega*, Vol. 34, No. 2, pp. 107-124.

26. Mangla S K, Luthra S, Mishra N *et al*. (2018), "Barriers to Effective Circular Supply Chain Management in a Developing Country Context", *Production Planning & Control*, Vol. 29, No. 6, pp. 551-569.

27. Meyer H (1991), "An Eight-Step Approach to Inventory Accuracy: A Case Study at Leviton Manufacturing Company", *Production and Inventory Management Journal*, Vol. 32, No. 2, p. 40.

28. Mohanty R P (1985), "Inventory Problems Under Multiple Constraints: Some Studies", *Engineering Costs and Production Economics*, Vol. 9, No. 4, pp. 355-367.

29. Muchaendepi W, Mbohwa C, Hamandishe T and Kanyepe J (2019), "Inventory Management and Performance of SMEs in the Manufacturing Sector of Harare", *Procedia Manufacturing*, Vol. 33, pp. 454-461.

30. Narayanapillai R (2013), "Inventory Management Practices in Small and Medium Machine Tool Enterprises in India: What Differentiate Between Enterprises?", *Asian Journal of Management*, Vol. 1, No. 01, pp. 05-11.

31. Natarajan R (1991), "Inventory Management-the Big Picture", *Production and Inventory Management Journal*, Vol. 32, No. 4, p. 29.

32. Ng S O E, Partington E C and Sculli D (1993), "A Computer System for Inventory Management of Lighting Products: A Case Study", *Computers in Industry*, Vol. 22, No. 1, pp. 71-79.

33. Orobia L A, Nakibuuka J, Bananuka J and Akisimire R (2020), "Inventory Management, Managerial Competence and Financial Performance of Small Businesses", *Journal of Accounting in Emerging Economies*, Vol. 10, No. 3, pp. 379-398.

34. Palange A and Dhatrak P (2021), "Lean Manufacturing a Vital Tool to Enhance Productivity in Manufacturing", *Materials Today: Proceedings*, Vol. 46, pp. 729-736.

35. Peel M J and Wilson N (1996), "Working Capital and Financial Management Practices in the Small Firm Sector", *International Small Business Journal*, Vol. 14, No. 2, pp. 52-68.

36. Pisano G and Teece D (1994), "The Dynamic Capabilities of Firms: An Introduction", *Industrial and Corporate Change*, Vol. 3, No. 3, pp. 537-556.

37. Porter M E (1998), "Clusters and the New Economics of Competition", *Harvard Business Review: Boston*, Vol. 76, No. 6, pp. 77-90.

38. Porter M E (1980), *Competitive Strategy*, Free Press, New York.

39. Rajeev N (2008), "Inventory Management in Small and Medium Enterprises: A Study of Machine Tool Enterprises in Bangalore", *Management Research News*, Vol. 31, No. 9, pp. 659-669.

40. Saratchandra M, Shrestha A and Murray P A (2022), "Building Knowledge Ambidexterity Using Cloud Computing: Longitudinal Case Studies of SMEs Experiences", *International Journal of Information Management*, Vol. 67, p. 102551.

41. Sharma A (2023), "Himachal Government Identifies 5 Sites for Setting Up Industrial Corridor in BBN Belt", *The Tribune*. Retrieved from https://www.tribuneindia.com/news/himachal/5-sites-identified-for-setting-up-of-industrial-corridor-in-bbn-belt-467497 Accessed 14 June 2023.

42. Sinha P, Kumar S and Chandra C (2023), "Strategies for Ensuring Required Service Level for Covid-19 Herd Immunity in Indian Vaccine Supply Chain", *European Journal of Operational Research*, Vol. 304, No. 1, pp. 339-352.

43. Sprague L G and Wacker J G (1996), "Macroeconomic Analyses of Inventories: Learning from Practice", *International Journal of Production Economics*, Vol. 45, Nos. 1-3, pp. 231-237.

44. Syntetos A A, Nikolopoulos K and Boylan J E (2010), "Judging the Judges Through Accuracy-Implication Metrics: The Case of Inventory Forecasting", *International Journal of Forecasting*, Vol. 26, No. 1, pp. 134-143.

45. Upadhye N, Deshmukh S G and Garg S (2010), "Lean Manufacturing System for Medium Size Manufacturing Enterprises: An Indian Case", *International Journal of Management Science and Engineering Management*, Vol. 5, No. 5, pp. 362-375.

46. Vergin R C (1998), "An Examination of Inventory Turnover in the Fortune 500 Industrial Companies", *Production and Inventory Management Journal*, Vol. 39, No. 1, p. 51.

47. Wernerfelt B (1984), "A Resource-Based View of the Firm", *Strategic Management Journal*, Vol. 5, No. 2, pp. 171-180.

48. Widyastuti W, Asandimitra N and Artanti Y (2018), "Inhibiting Factors of Inventory Management: Study on Food and Beverage Micro Small and Medium Enterprises", *International Review of Management and Marketing*, Vol. 8, No. 1, pp. 64-67.

Appendix

Questionnaire

1. How often do you take inventory in your business?	
2. How many SKUs do you have as inventory?	
3. At what inventory level, do you reorder?	
4. What is your lead time for receiving inventory from suppliers?	
5. Our inventory levels are frequently too high, leading to excess carrying costs.	
6. Our inventory management system helps us to avoid stockouts.	
7. We regularly conduct physical inventory counts to ensure that our records are accurate.	
8. Our inventory carrying costs are reasonable.	
9. We maintain safety stock levels to ensure that we can meet unexpected demand.	
10. What methods do you use to track your inventory levels?	
11. What is your approach to managing excess inventory?	
12. What steps do you take to ensure accurate inventory counts?	
13. How do you handle stockouts or shortages?	
14. What is your approach to inventory forecasting?	
15. How do you manage inventory storage and organization?	
16. Which concept of inventory system is followed?	
17. Which factors mostly affect your inventory practices?	

Appendix (Cont.)

18. Which method of procurement you follow for ordering inventory?	
19. What are the factors that cause delay in controlling inventory practices in your business?	
20. What are the most common problems you have faced in Baddi industrial area while managing inventory in your business?	
21. How do you manage your finished goods inventory?	

Reference # 07J-2024-05-03-01

Copyright of IUP Journal of Operations Management is the property of IUP Publications and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.